

I. Benzene Structure and Resonance

Normal State: In its natural state, benzene possesses delocalized π electrons that form an electronic cloud above and below the ring, rather than being fixed in static double bonds.

Resonance Requirements: For resonance to occur, a specific sequence must exist: double-single-double bonds, a double bond separated by a single bond from a lone pair, a negative charge, or a positive charge.

Electron Movement: During resonance, the movement of electrons from one atom to another leaves a positive charge on the atom that lost them. Carbon atoms cannot remain negatively charged for long due to their lower electronegativity compared to atoms like nitrogen; therefore, they continue the resonance cycle to redistribute the charge.

II. Substituent Effects: Directing and Activating Groups

Electron-Donating Groups (Activators): Groups like the amino group ($-\text{NH}_2$) have lone pairs that participate in resonance with the ring. They increase electron density specifically at the ortho and para positions.

Electron-Withdrawing Groups (Deactivators): Groups like the nitro group ($-\text{NO}_2$) pull electrons away from the ring. This creates positive charges at the ortho and para positions, so electrophiles attack the meta position instead.

The Halogen Exception: Halogens are deactivators but direct to ortho/para positions because they possess lone pairs that can participate in resonance.

III. Aromaticity Requirements

A compound is classified as aromatic if it is cyclic, planar, and fully conjugated (all atoms sp^2 hybridized).

Huckel's Rule: It must have $(4n+2)$ π electrons, where n is a whole number.

Non-aromatic: Compounds that fail any of the first three conditions.

Anti-aromatic: Compounds that meet the first three conditions but have $4n$ electrons; these are very unstable.

IV. Electrophilic Aromatic Substitution (SEAr) Reactions

Nitration: Uses concentrated H_2SO_4 and HNO_3 to generate NO_2^+ .

Sulfonation: Introduces the $-\text{SO}_3\text{H}$ group.

Halogenation: Requires a halogen and a Lewis acid catalyst.

Friedel-Crafts Alkylation: Introduces an alkyl group using an alkyl halide and a Lewis acid catalyst.

Friedel-Crafts Acylation: Introduces an acyl group using an acid chloride or acetic anhydride.

V. Critical Limitations and Exceptions in Friedel-Crafts

Deactivated Rings: Do not react if strong electron-withdrawing groups are present.

Strong Activators with Lone Pairs: Groups like -NH_2 or -OH react with AlCl_3 , reducing activation.

Rearrangement: Alkylation can rearrange; acylation followed by Clemmensen reduction avoids this.

VI. Nucleophilic Aromatic Substitution ($\text{S}_{\text{N}}\text{Ar}$)

Aryl halides do not undergo $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$ reactions.

Addition-Elimination: Stabilized by electron-withdrawing groups.

Elimination-Addition: Proceeds via benzyne intermediate using strong base like NaNH_2 .

VII. Key Nomenclature and Intermediates

Phenyl vs. Benzyl: Phenyl is a benzene substituent; Benzyl includes a CH_2 group.

Sigma Complex (Arenium Ion): Positively charged intermediate in substitution.

Allyl vs. Vinyl: Allyl has three carbons with a double bond; Vinyl has two.
